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The Impact of Attentional Resources on Emotional Expressions Revealed by Event-Related Potentials

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ABSTRACT

The prioritized processing for emotions has been found to modulate the N170, Early Posterior Negativity (EPN), and Late Positive Potential (LPP) within the processing of facial expressions. However, existing evidence on how available attentional resources affect the emotional modulation of these ERP components remains inconsistent. To investigate this issue, we precisely controlled the perceptual load of the first target (T1) in a dual-target rapid serial visual presentation (RSVP) task, aiming to modulate the attentional resources available for processing emotional expressions (happy, fearful, neutral) at the second target (T2). Our results revealed the main effects of emotion in the 180–280 ms and 400–800 ms intervals, which were consistent with the early-stage processing of emotional expressions, as well as the elaborate, late-stage processing. During 320–400 ms, the amplitudes were more negative for high compared to low load conditions, representing a bottleneck in visual processing. The interactions between emotion and perceptual load were observed for the late LPP interval (600–800 ms). Specifically, the processing of happy expressions was modulated by perceptual load levels, whereas fearful expressions exhibited robust prioritized processing even under limited attentional resources. Overall, the present study revealed that the early-stage processing of emotional expressions operates automatically due to their high biological salience. In contrast, late-stage processing relies on the available attentional resources, with fearful expressions requiring minimal attentional resources due to their evolutionary significance.

1 | Introduction

Perceptual load theory posits that attentional resources are fully allocated to task-relevant stimuli, leaving no resources for distractors under high load conditions. In contrast, low load conditions allow for the spillover of attentional resources, enabling the processing of distractors (Lavie 2005; Lavie et al. 2004). A large body of studies has demonstrated that emotional stimuli attract attention faster than neutral ones (Vuilleumier 2005; Zsidó et al. 2023), providing an advantage in perception (Gupta et al. 2019; Pool et al. 2016). However, there was ongoing debate about whether and when this emotional superiority effect was influenced by perceptual load levels (Müller-Bardorff et al. 2016; Schindler, Caldarone, et al. 2020; Schindler, Tirloni, et al. 2021).

Due to their high temporal resolution, event-related potentials (ERPs) serve as an effective method for examining at which stage perceptual load can reduce or eliminate the prioritized processing of emotional expressions. Emotional expressions have been found to enhance several ERP components, including the N170, Early Posterior Negativity (EPN), and Late Positive Potential (LPP). More specifically, the N170 component, a structural encoding component, is typically larger for faces than for objects (Eimer 2011). Some studies report enhanced N170

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amplitudes for fearful expressions (Luo et al. 2010; Schindler et al. 2019), suggesting preferential processing of threats. Others suggest that the N170 component responds to emotional facial expressions without robust differences between emotional valences (Hinojosa et al. 2015). Evidence from the spatial cueing paradigm indicates that emotional modulation of the N170 is affected by perceptual load (Eimer et al. 2003). Müller-Bardorff et al. (2016) manipulated perceptual load through the relative length of two bars superimposed over faces. They found that angry expressions elicited stronger amplitudes than happy ones during low perceptual load, while this difference was not evident when perceptual load was high. However, Li et al. (2024) also employed the bar length detection task, and Schindler, Bruchmann, et al. (2021) employed the letter detection task reporting that perceptual load does not significantly influence the emotional modulation of the N170 component, suggesting that emotional facial expressions are processed automatically at early stages.

The subsequent EPN reflects emotional modulation and the engagement of early selective attention (Schindler and Bublatzky 2020; Schupp et al. 2006). A larger negativity for emotional expressions than neutral ones was observed in many studies, but the differences across emotional valences did not reach significance (Calvo and Beltrán 2013; Rellecke et al. 2012). This highlights the important contribution of emotional arousal to EPN modulation (Schindler, Tirloni, et al. 2021). Several studies have found that perceptual distraction tasks do not significantly modulate the enhanced amplitudes for emotional expressions (Itier and Neath-Tavares 2017; Valdés-Conroy et al. 2014). However, recent studies found that the emotional modulation of EPN primarily emerges when attention is directed to the emotional information within faces (Schindler, Bruchmann, et al. 2020). Considering the impact of perceptual load, Müller-Bardorff et al. (2016) observed that the EPN deflections were enhanced by angry expressions compared to happy expressions under low load rather than under high load conditions. Nevertheless, Li et al. (2024) employed the same task and failed to find the interaction between perceptual load and emotional valences.

Finally, the LPP is considered a marker of extended attentional engagement and thorough assessment of emotionally or motivationally significant stimuli (Schindler and Bublatzky 2020; Schupp and Kirmse 2021). Given the temporally extended nature of the LPP component, some research has segmented it into an early phase (approximately 400–600 ms) and a later phase (beyond 600 ms). The early portion of the LPP primarily reflects the initial orienting of attention toward stimuli with motivational significance (Olofsson et al. 2008; Schupp et al. 2006; Uusberg et al. 2013). Schindler and Straube (2020) investigated the interplay between emotion and task relevance, reporting a significant interaction during the early LPP (400–600 ms), while noninteracting effects characterized later stages. This finding has been viewed as evidence that the early LPP integrates emotional input from the amygdala with top-down task-related modulation from frontoparietal regions (Pourtois et al. 2013). In contrast, the late LPP is believed to index more elaborative processing associated with the meaning and significance of stimuli, and is more strongly influenced by the availability of attention resources (Olofsson et al. 2008; Schupp et al. 2006; Uusberg et al. 2013). Findings regarding how perceptual load affects the emotional modulations of the LPP are mixed. Some researchers indicate that the emotional modulation of the LPP is diminished merely under high load conditions (Holmes et al. 2009; Roberge et al. 2019), while other studies report no significant LPP emotional response, regardless of perceptual load levels (Li et al. 2024; Müller-Bardorff et al. 2016; Schindler, Bruchmann, et al. 2021).

Simultaneous presentation of targets and distractors in previous studies may have obscured the allocation of attentional resources to emotional expressions, contributing to inconclusive results (Li et al. 2023; Li et al. 2024; Ye et al. 2011). According to the neural theory of visual attention (Giesbrecht et al. 2014), perceptual load theory assumes a strict order of attention allocation. That is, attentional resources are preferentially allocated to task-relevant information, with any residual capacity subsequently distributed to task-irrelevant stimuli. Accordingly, there should be no effect of task-irrelevant objects on task performance. However, distractors, especially emotional distractors, can capture attention (Carlson and Mujica-Parodi 2015; Carlson and Reinke 2014; Gupta 2019; Pool et al. 2016), which could influence the task outcomes and obscure the allocation of attentional resources to facial expressions (Forster and Lavie 2011; Li et al. 2024). Moreover, previous studies cannot rule out the possibility that the simultaneously presented target and distractor stimuli may be processed in parallel at the initial stage, with the actual selection occurring at a later stage (Benoni 2018; Giesbrecht et al. 2014). Two recent studies have considered the temporal aspect of perceptual load (Schindler, Bruchmann, et al. 2020; Schindler, Tirloni, et al. 2021). In both studies, high perceptual load was manipulated by reducing the perceptibility of stimulus differences—either through very subtle grayscale variations between the target (RGB 80) and distractors (RGB 90) (Schindler, Caldarone, et al. 2020) or via adaptive adjustments of color contrast (Schindler, Tirloni, et al. 2021). Consequently, the extremely low accuracy (13.13%) and the 51% hit rate, respectively, observed under high load conditions may reflect sensory discrimination limitations rather than attentional resource depletion. Moreover, both studies focused exclusively on early ERP components. It is essential to manipulate the temporal framework of perceptual load through strictly controlling attentional resources and to consider its impact on all stages of emotional face processing.

The dual-target RSVP paradigm is a widely used temporal design that manipulates the interval between two targets to regulate the attentional resources devoted to the second target (T2) (Li et al. 2023; Pecchinenda et al. 2020; Raymond et al. 1992). In this task, stimuli are rapidly presented at a single location (typically 6–20 items per second), and participants must identify two target items while ignoring intervening distractors (Dux and Marois 2009). When the temporal lag between the two targets is 200–500 ms, accurately identifying T1 leads to a significant reduction in T2 discrimination accuracy, a phenomenon known as the attentional blink (AB) (Raymond et al. 1992). Converged evidence shows that the AB is attenuated when emotional stimuli, such as words (Anderson 2005; Zhang et al. 2014), eye regions (Li et al. 2022; Wang et al. 2022), and facial expressions (Luo et al. 2010; Maratos et al. 2008), are presented at T2. Emotionally arousing stimuli appear to bypass the attentional

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constraints imposed by T1 processing, enabling postperceptual-level processing. Moreover, Giesbrecht et al. (2007) varied the perceptual load levels of the T1 task, with T2 being a word either semantically related or unrelated to a context word shown at the start of each trial. Results showed that the semantically relevant N400 component was absent at high perceptual load conditions during the AB period. This finding indicates that the postperceptual processing of T2 was regulated by the perceptual load levels at T1 (Forster and Lavie 2011). Therefore, the temporal aspect of perceptual load using a dual-target RSVP task appears well suited for examining how attentional resources influence different stages of emotional expression processing.

Another factor contributing to the contradictory evidence in this field is the ERP analysis methodology used across studies. The traditional spatiotemporal averaging approach often selects the electrodes and time points through data visualization before statistical analysis. This practice leads to implicit multiple comparisons, significantly inflating the Type I error rate (Luck and Gaspelin 2017; McCrackin and Itier 2021). Analyzing a priori time points and electrode sites can control Type I errors, but it also reduces the power of detecting unexpected yet real effects, thereby increasing the likelihood of Type II errors (Fields and Kuperberg 2020). In contrast, the recently developed mass univariate analysis applies multiple comparison corrections during hypothesis testing, helping to control Type I error (Fields and Kuperberg 2020; Groppe et al. 2011; Luck and Gaspelin 2017). Additionally, the exploratory analysis test, which examines all electrodes and time points, can uncover unpredicted but genuine effects, thereby reducing Type II errors (McCrackin and Itier 2021). The mass univariate analysis offers slightly greater statistical power than traditional ERP approaches in hypothesis-driven investigations. During exploratory analysis, their power declines gradually, but less dramatically than traditional approaches (Fields and Kuperberg 2020; Groppe et al. 2011). Therefore, the present study utilized the mass univariate analysis to conduct both exploratory and hypothesis-driven analyses.

In this study, we utilized the modified dual-target RSVP task combined with mass univariate analysis to examine whether, and at which processing stage, the prioritized processing of emotional expressions is modulated by attentional resources manipulated via T1 load levels. Based on previous literature (Giesbrecht et al. 2007; Park et al. 2014), the present study controlled the perceptual load levels through the letter detection task at T1. Participants were requested to respond to neutral, fearful, and happy expressions at T2. According to a previous study (Pecchinenda et al. 2020), the recognition accuracy of T2 was the lowest when T1 was a neutral stimulus and T2 appeared at lag2, reflecting minimal available attentional resources at this point. Based on this, we displayed the T2 stimulus at a fixed lag2 position (stimulus onset asynchrony, SOA = 232 ms). Based on previous studies (Hudson et al. 2021; Itier and Neath-Tavares 2017; Schindler, Caldarone, et al. 2020; Schindler, Tirloni, et al. 2021), we predicted that fearful and happy expressions would elicit enhanced N170 amplitudes compared to neutral expressions, and that these emotional effects would be independent of perceptual load. Considering that the EPN reflects emotional modulation and the engagement of early selective attention (Müller-Bardorff et al. 2016; Schupp et al. 2006), we expected the effect of perceptual load to emerge at the EPN time window, and the EPN emotional effects should be observed. Furthermore, given the sensitivity of the late LPP to attentional capacity (Olofsson et al. 2008; Schupp et al. 2006; Uusberg et al. 2013) and evidence that high perceptual load can disrupt post-perceptual processing of T2 stimuli (Giesbrecht et al. 2007), we anticipated that the modulation of emotional expression processing by perceptual load would also be evident during this later time window.

2 | Method

2.1 | Participants

Thirty college students (18 females; mean age = 21.23 years, SD = 1.89) from Liaoning Normal University participated in the study. A priori power analysis using G*Power 3.1 (Faul et al. 2007) confirmed that this sample size was sufficient to detect a medium effect size (Cohen's $f = 0.25$) with 95% power at $\alpha = 0.05$. All participants were right-handed with normal or corrected vision, and denied any history of neuropsychological or psychiatric disorders. Ethical approval was obtained from the local ethics committee.

2.2 | Stimuli

In this experiment, a total of 32 facial images (20 neutral, six happy, and six fearful) were selected from the Chinese Facial Affective Picture System (CFAPS; Gong et al. 2011) as the original pictures (half female). Each image was cropped into an elliptical shape using Adobe Photoshop 8.0, resulting in a final size of 276 × 340 pixels. When viewed from a distance of 70 cm on a 19-in. monitor, the stimuli subtended a visual angle of approximately 6.0° (horizontal) × 7.4° (vertical). Low-level physical features, including brightness and contrast, were equated using the SHINE toolbox (Willenbockel et al. 2010). Eighteen upright images (six happy, six neutral, and six fearful) were used as T2 stimuli (Figure 1c). Additionally, 12 inverted neutral expressions were used as distractors. The remaining two upright neutral expressions (one female) were overlaid with a string of five white letters centered on the facial midpoint (Park et al. 2014). In the high load conditions, each letter string contained one target letter (X or Z) and four nontarget letters (H, K, N, F), arranged randomly. In the low load conditions, the letter string comprised five identical letters (either all Xs or all Zs) (Figure 1b).

Before the experiment, 28 participants (17 females, mean age = 21.82 years, SD = 1.74) rated the valence and arousal of the T2 upright faces using a 9-point Likert scale. Valence ratings differed significantly among the three facial expressions [F (2, 54) = 350.88, $p < 0.001$, $\eta_p^2 = 0.93$]. Follow-up test revealed that happy (M ± SD, 7.79 ± 0.76, $p < 0.001$) and neutral expressions (4.74 ± 0.85, $p < 0.001$) were rated as more pleasant than fearful expressions (2.29 ± 0.79). And the happy expressions were more pleasant compared to the neutral expressions ($p < 0.001$). Arousal ratings also differed significantly [F (2, 54) = 51.99, $p < 0.001$, $\eta_p^2 = 0.66$]. Both happy (M ± SD, 6.64 ± 1.70, $p < 0.001$) and fearful (6.74 ± 1.42, $p < 0.001$) expressions were rated as significantly more arousing than neutral expressions (3.52 ± 1.27). The difference in arousal between happy and fearful expressions was not significant ($p = 1.000$).

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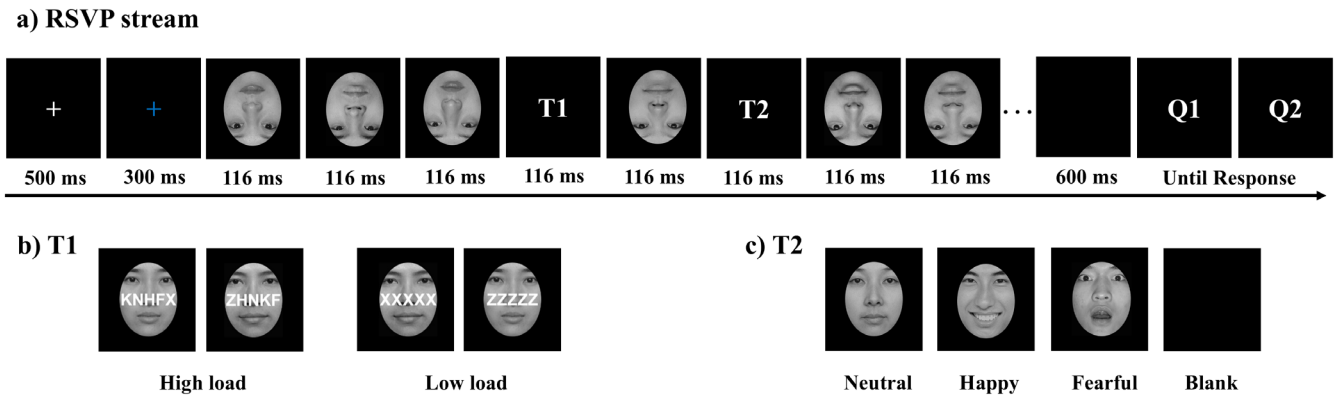


FIGURE 1 | (a) An example trial in which T1 appeared in the fourth position is shown. Each image in the RSVP stream was presented for 116 ms. After the RSVP stream presentation, participants answered two questions. (b) Sample face stimuli used at T1. (c) Sample face stimuli used at the T2.

2.3 | Procedure

The experiment was executed in a soundproofed, mildly lit room. Participants were seated 70 cm from a 19-in. computer monitor. The display featured a resolution of 1440 × 900 pixels and operated at a refresh rate of 60 Hz. The design included six conditions: three emotional valences (fearful, happy, neutral) × two perceptual load levels (high, low). The experiment was implemented via E-Prime 2.0 software (Psychology Software Tools Inc., Pittsburgh, PA, USA).

Each trial started with a 500 ms white fixation cross, followed by a 300 ms blue fixation cross signaling that the stimulus would be presented soon (Figure 1a). A sequence of 12 distractors (inverted neutral expressions) and two targets (T1 and T2) then successively appeared, each displayed for 116 ms. To reduce anticipatory effects (Akyürek and Hommel 2006; Raymond et al. 1992), T1 was presented randomly at either the fourth or sixth position. T2 followed at lag 2 (SOA = 232 ms). The control condition was also implemented to obtain pure ERPs related to T2 (Li et al. 2023; Luo et al. 2010; Vogel et al. 1998). In the control condition, a black screen rather than faces was presented at the T2 position. After the stream, a 600 ms blank screen appeared. Then, two response questions followed. Q1: Which letter appears in the string, X or Z? Participants pressed "1" for "X," "2" for Z. Q2: What is the expression of the second upright face? Participants should enter "1" for neutral, "2" for happy, "3" for fearful, or "0" for "absence of the T2." Participants were instructed to respond as accurately as possible, with no time limit. A 600 ms black interval preceded the next trial.

Participants completed 16 practice trials before the main experiment, which comprised six randomized blocks (three high load, three low load). Within each block, the emotional valence conditions were counterbalanced. Each block contained 96 trials, resulting in 576 trials in total. The entire experiment lasted approximately 60 min.

2.4 | EEG Data Recording and Preprocessing

EEG data were recorded from 64 scalp electrodes mounted on elastic caps following the extended 10–20 system (Brain Products, Munich, Germany). The data were sampled at 1000 Hz with FCz serving as the online reference, and all electrode impedances were kept under 5 kΩ. An electrode was positioned 10 mm beneath the right eye to record the vertical electrooculogram (VEOG). A band-pass filter of 0.01 to 100 Hz was applied during the online recording.

Data preprocessing was performed offline in MATLAB R2018b, utilizing the EEGLAB (version 2020.0; Delorme and Makeig 2004) and ERPLAB (version 8.01; Lopez-Calderon and Luck 2014) toolboxes. The steps included: applying a 0.01–0.30 Hz offline band-pass filter; interpolating noisy channels using multivariate local weighted regression in EEGLAB; re-referencing to the average; and segmenting data into epochs from 200 ms before T2 onset (serving as the baseline) to 800 ms after it. Only trials with correct T1 and T2 responses were retained. Independent Component Analysis was conducted to remove ocular and channel artifacts, generating 30 components per participant. On average, 2.63 (SD = 1.61) components were removed per participant. Finally, trials with peak-to-peak amplitudes exceeding ±80 μV were automatically rejected on all channels except for the frontal sites, including Fpz, Fp1, Fp2, AF3, AF4, AF7, AF8. The mean number of remaining trials per condition was as follows (M ± SE): high load neutral, 55.50 ± 1.76; high load happy, 55.40 ± 1.75; high load fearful, 56.73 ± 1.42; high load control, 57.27 ± 1.76; low load neutral, 63.20 ± 1.51; low load happy, 62.10 ± 1.74; low load fearful, 62.33 ± 1.60; and low load control, 63.27 ± 1.68. Finally, we applied the subtraction method (Zivony and Lamy 2022) to isolate the neural activity specifically elicited by the T2 emotional stimuli. In this approach, neural responses in each of the six experimental conditions were subtracted from their corresponding control condition.

2.5 | Data Analysis

2.5.1 | Behavioral Analysis

Behavioral data were analyzed using SPSS software (version 25.0, SPSS Inc., Chicago, IL). A paired-sample *t*-test was used to investigate the effect of perceptual load on T1 performance. Given the uncertainty surrounding T2 errors in incorrect T1 trials (Li et al. 2023), T2 accuracy was computed

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only for correct T1 trials. To examine the effects on T2 accuracy, we performed a two-way repeated-measures ANOVA that included perceptual load (high, low) and emotional valence (happy, neutral, fearful) as within-subject factors. T1 and T2 performance were also evaluated using inverse efficiency scores, which account for both accuracy and reaction time (Vandierendonck 2017; see Supporting Information). Greenhouse–Geisser corrections were applied when sphericity was violated (Mauchly's test). Bonferroni adjustments were used for post hoc comparisons where appropriate. Effect sizes were reported using Cohen's d and partial eta-squared (η_p^2), with significance set at $p < 0.05$.

2.5.2 | Factorial Mass Univariate Analysis

Factorial Mass Univariate Toolbox (FMUT; Fields and Kuperberg 2020) was used to analyze the EEG data. An exploratory ANOVA was initially conducted across all electrodes and time points from 0 to 800 ms. To better visualize early emotional effects, we additionally plotted scalp topographies and difference topographies from 0 to 300 ms in 30-ms increments (see Figures S1 and S2). Based on the scalp and difference topographies and exploratory results, and supported by literature on early component delays in RSVP tasks (Luo et al. 2010; Tian et al. 2018; Zhang et al. 2014), we selected two priori intervals (180–280 and 320–400 ms) at occipitotemporal electrodes (P7, P8, PO7, PO8). For later ERP components, analyses focused on the early (400–600 ms) and late LPP (600–800 ms) at central–parietal sites (C1, C3, Cz, C2, C4, CP1, CP3, CPz, CP2, CP4, P3, Pz, P4). For each ANOVA, the within-subjects factors were perceptual load (high, low) and emotional valence (happy, fearful, neutral), with an alpha level set at 0.05. Follow-up comparisons were Bonferroni-corrected (e.g., $\alpha = 0.0083$ for six pairwise tests). Furthermore, 100,000 permutations were applied to every data point in the ANOVAs.

3 | Results

3.1 | Behavioral Results

A paired-sample t-test on T1 accuracy revealed significantly better performance under the low load relative to the high load condition (99% vs. 92%, $t(29) = 8.35$, $p < 0.001$, Cohen's $d = 1.52$). Consistent with previous studies (Giesbrecht et al. 2007; Schindler, Caldarone, et al. 2020), this result indicated that the load manipulation in the present study was effective. For T2 accuracy, a two-way ANOVA exhibited a significant main effect for emotional valence [$F(2, 58) = 30.44$, $p < 0.001$, $\eta_p^2 = 0.51$]. Post hoc tests observed higher recognition accuracy for the happy (93%, $p < 0.001$) and the fearful (94%, $p < 0.001$) compared to the neutral expressions (86%). Happy and fearful expressions did not differ significantly ($p = 0.128$). A significant main effect of perceptual load was also observed [$F(1, 29) = 66.53$, $p < 0.001$, $\eta_p^2 = 0.70$]. Participants showed a higher accuracy in the low load condition (96%) compared to the high load condition (87%, $p < 0.001$). Moreover, a significant interaction was observed between perceptual load and emotional valence [$F(2, 58) = 12.10$, $p < 0.001$, $\eta_p^2 = 0.29$]. Under high perceptual load, accuracy was higher for fearful (91%, $p < 0.001$) and happy expressions (89%, $p < 0.001$) compared to neutral expressions (80%), with no significant difference between fearful and happy expressions ($p = 0.364$). Similarly, in the low load condition, accuracy was higher for fearful (98%, $p = 0.001$) and happy expressions (97%, $p = 0.017$) than for neutral expressions (93%), and the difference between fearful and happy expressions remained nonsignificant ($p = 0.229$).

3.2 | ERP Results

3.2.1 | Exploratory Analysis (All Electrodes; 0–800 ms)

A widespread main effect of emotional valence (from 132 to 798 ms) was observed in the omnibus ANOVA (Figure 2a; Table 1). This cluster spanned a broad range of electrodes, including frontal, parietal, central, occipitotemporal, and parieto-occipital sites. The maximal effect occurred at C1 at 476 ms [$F(2, 58) = 100.51$, $p = 0.00001$]. Two distinct clusters primarily accounted for the significant main effect of perceptual load (Figure 2b). The first cluster from 274 to 388 ms peaked at F3 at 306 ms [$F(1, 29) = 15.99$, $p = 0.04511$]. The second cluster, ranging from 280 to 388 ms, peaked at P6 at 304 ms [$F(1, 29) = 20.67$, $p = 0.02899$] and involved electrodes over parietal, occipital, and occipitotemporal sites. Furthermore, a significant interaction between emotional valence and perceptual load was also observed, peaking at CPz at 616 ms [$F(2, 58) = 12.71$, $p = 0.00338$] (Figure 2c). This cluster ranged from 436 to 768 ms, including distributed electrodes of center, parietal, and frontal sites, which encompassed the LPP component.

3.2.2 | Prior Analysis (Occipito-Temporal Electrodes; 180–280 ms)

A significant main effect of emotional valence emerged in two clusters (Figure 3; Table 2). Follow-up ANOVAs indicated that amplitude differences between fearful and neutral expressions were observed in two clusters (Figure 3a). The first cluster peaked at P7 at 278 ms [$F(1, 29) = 34.95$, $p = 0.00003$] and the second at PO8 at 230 ms [$F(1, 29) = 53.36$, $p = 0.00001$]. Happy expressions elicited more negative amplitudes than neutral expressions (Figure 3b), with the effect peaking at electrode P7 around 280 ms [$F(1, 29) = 57.63$, $p = 0.00003$] and PO8 at 226 ms [$F(1, 29) = 67.98$, $p = 0.00001$]. Additionally, negative amplitudes were observed for happy compared with fearful expressions (Figure 3c), peaking at PO8 at 190 ms [$F(1, 29) = 25.42$, $p = 0.00193$]. No significant effects were observed for perceptual load ($ps > 0.347$) or its interaction with emotional valence ($ps > 0.126$).

3.2.3 | Prior Analysis (Occipito-Temporal Electrodes; 320–400 ms)

A significant main effect of perceptual load was observed in the right hemisphere (Figure 4; Table 2). The effect was characterized by more negative amplitudes for high load than for low load conditions, peaking at PO8 at 320 ms [$F(1, 29) = 13.01$, $p = 0.0036$]. No significant main effect of emotional valence ($ps > 0.193$) or interaction with perceptual load ($ps > 0.061$) was observed.

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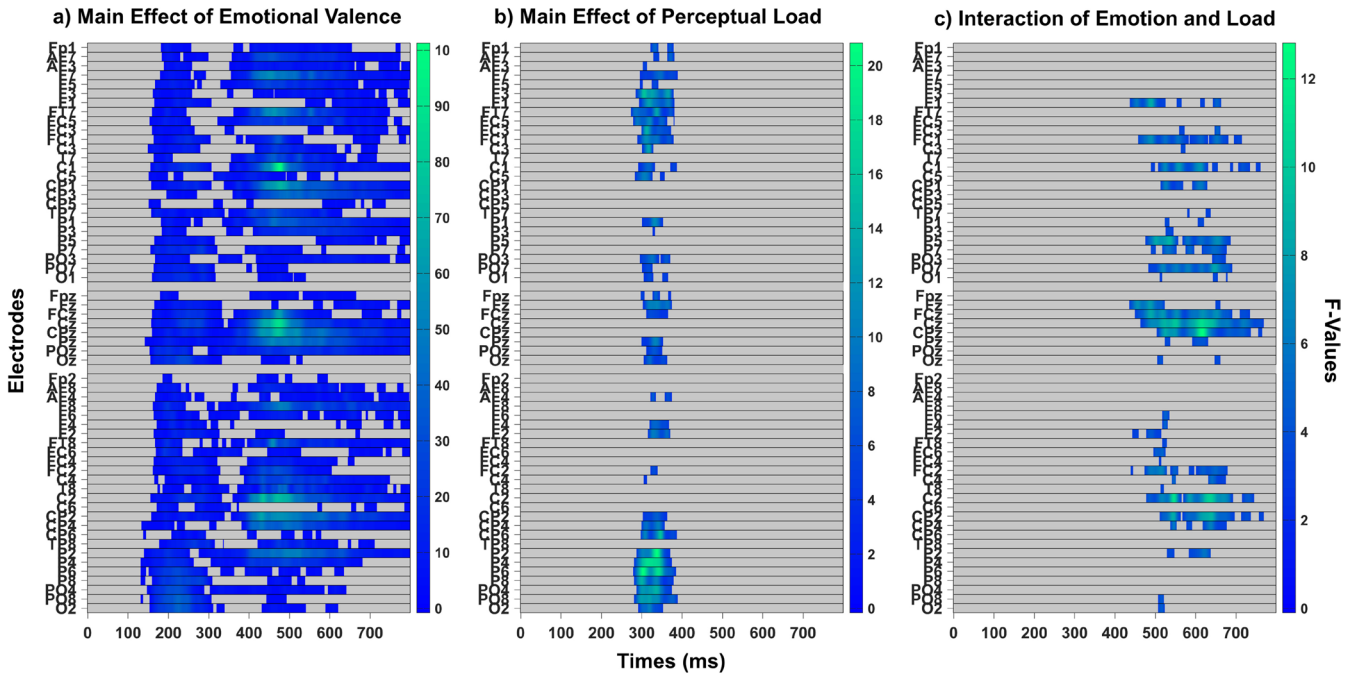


FIGURE 2 | Exploratory analysis across all electrodes (0–800 ms), corrected for multiple comparisons using the Permutation-Based Cluster Mass technique ($p < 0.05$). Panels show: (a) the main effect of emotional valence, (b) the main effect of perceptual load, and (c) the interaction of perceptual load and emotional valence.

TABLE 1 | Summary of the main effect of exploratory analysis ($p < 0.05$).

Test	Timing	Electrodes	Peak
Emotional valence	132–798 ms	Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T7, T8, P7, P8, Fz, Cz, Pz, Oz, FC1, FC2, CP1, CP2, FC5, FC6, CP5, CP6, FCz, F1, F2, C1, C2, P1, P2, AF3, AF4, FC3, FC4, CP3, CP4, PO3, PO4, F5, F6, C5, C6, P5, P6, AF7, AF8, FT7, FT8, TP7, TP8, PO7, PO8, Fpz, CPz, POz	C1 at 476 ms [$F(2, 58) = 100.51$, $p = 0.00001$]
Perceptual load	274–388 ms	Fp1, F3, F4, C3, F7, Fz, FC1, FC2, FC5, FCz, F1, F2, C1, AF3, AF4, FC3, F5, C5, AF7, FT7, Fpz	F3 at 306 ms [$F(1, 29) = 15.99$, $p = 0.04511$]
	280–388 ms	C4, P3, P4, O1, O2, P8, Pz, Oz, CP2, CP6, P1, P2, CP4, PO3, PO4, P6, PO7, PO8, POz	P6 at 304 ms [$F(1, 29) = 20.67$, $p = 0.02899$]
Emotion × Load	436–768 ms	F4, C3, C4, P3, O1, O2, T8, P7, Fz, Cz, Pz, Oz, FC1, FC2, CP1, CP2, FC6, FCz, F1, F2, C1, C2, P1, P2, FC3, FC4, CP4, PO3, F6, P5, FT8, TP7, PO7, PO8, CPz	CPz at 616 ms [$F(2, 58) = 12.71$, $p = 0.00338$]

Note: Listed are the time windows, electrode sites, and corresponding peak *F*- and *p* values reflecting significant effects observed in the analysis.

3.2.4 | Prior Analysis (Centro-Parietal; 400–600 ms)

There was a main effect of emotional valence (Figure 5; Table 2). Follow-up ANOVAs revealed significant amplitude differences between fearful and neutral expressions (Figure 5a), with the peak effect at C1 at 478 ms [$F(1, 29) = 136.11$, $p = 0.00001$]; happy and neutral expressions (Figure 5b), peaking at C2 at 432 ms [$F(1, 29) = 89.15$, $p = 0.00001$]; fearful and happy expressions (Figure 5c), with the peak effect at C1 at 476 ms [$F(1, 29) = 81.31$, $p = 0.00001$].

Additionally, a significant interaction between perceptual load and emotional valence was also observed (Figure 6; Table 2). Subsequent ANOVAs revealed similar response patterns under both high and low perceptual load conditions. Under high load conditions, enhanced amplitudes were observed for fearful relative to neutral expressions (Figure 6a), peaking at Cz at 472 ms [$F(1, 29) = 164.11$, $p = 0.00001$]; for happy relative to neutral expressions (Figure 6b), peaking at C1 at 484 ms [$F(1, 29) = 61.29$, $p = 0.00001$]; and for fearful compared to happy expressions (Figure 6c), peaking at CP1 at 522 ms

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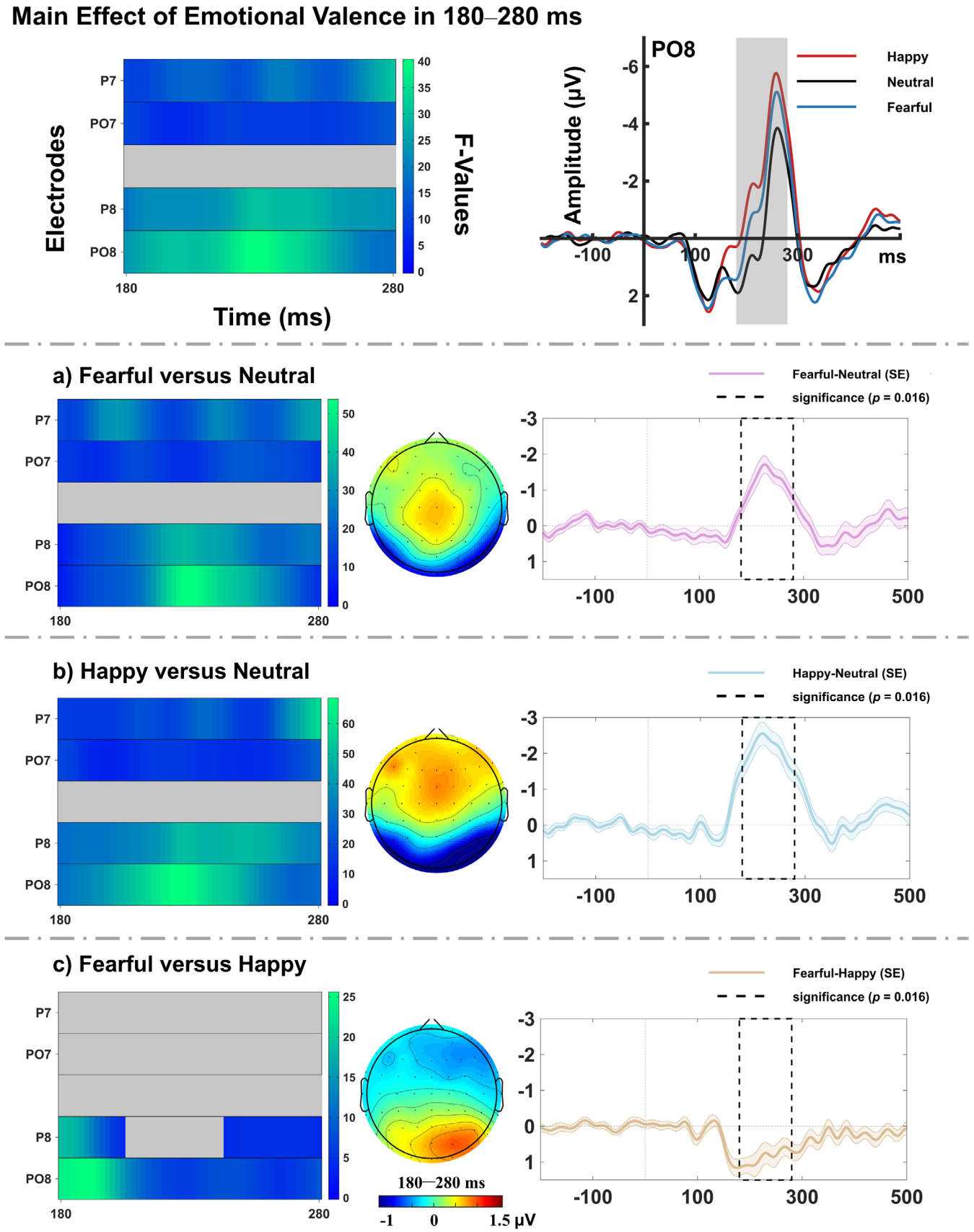


FIGURE 3 | Emotional valence main effect in the 180–280 ms time window. Grand average ERP waveforms are shown at the representative electrode (PO8) for happy (red), fearful (blue), and neutral (black) conditions. Statistical comparisons were conducted for (a) fearful versus neutral, (b) happy versus neutral, and (c) fearful versus happy conditions. The corresponding difference topographies and difference waveforms at PO8 was displayed at the middle panel and the right panel, respectively.

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TABLE 2 | Summary of the main effect of prior analysis ($p < 0.05$).

Priori analyses	Test	Timing	Electrodes	Peak
180–280 ms	Emotional valence	180–280 ms	PO7, P7	P7 at 280 ms [$F(2, 58) = 30.41$, $p = 0.00001$]
		180–280 ms	PO8, P8	PO8 at 230 ms [$F(2, 58) = 40.10$, $p = 0.00001$]
	Perceptual load	N/A	N/A	N/A
	Emotion × load	N/A	N/A	N/A
320–400 ms	Emotional valence	N/A	N/A	N/A
	Perceptual load	320–388 ms	PO8, P8	PO8 at 320 ms [$F(1, 29) = 13.01$, $p = 0.0036$]
	Emotion × load	N/A	N/A	N/A
400–600 ms	Emotional valence	400–600 ms	C1, C2, Cz, CP1, CP2, CPz, P1, Pz, P2	C1 at 476 ms [$F(2, 58) = 100.51$, $p = 0.00001$]
	Perceptual load	N/A	N/A	N/A
	Emotion × load	464–600 ms	C1, C2, Cz, CP1, CP2, CPz, P1, Pz, P2	C2 at 546 ms [$F(2, 58) = 11.20$, $p = 0.00444$]
600–800 ms	Emotional valence	600–798 ms	C1, C2, Cz, CP1, CP2, CPz, P1, Pz, P2	CP2 at 632 ms [$F(2, 58) = 37.70$, $p = 0.00001$]
	Perceptual load	N/A	N/A	N/A
	Emotion × load	600–768 ms	C1, C2, Cz, CP1, CP2, CPz, P1, Pz, P2	CPz at 616 ms [$F(2, 58) = 12.71$, $p = 0.00299$]

Note: This table lists the significant time windows, associated electrodes, and corresponding peak *F*- and *p*-values from the analyses.

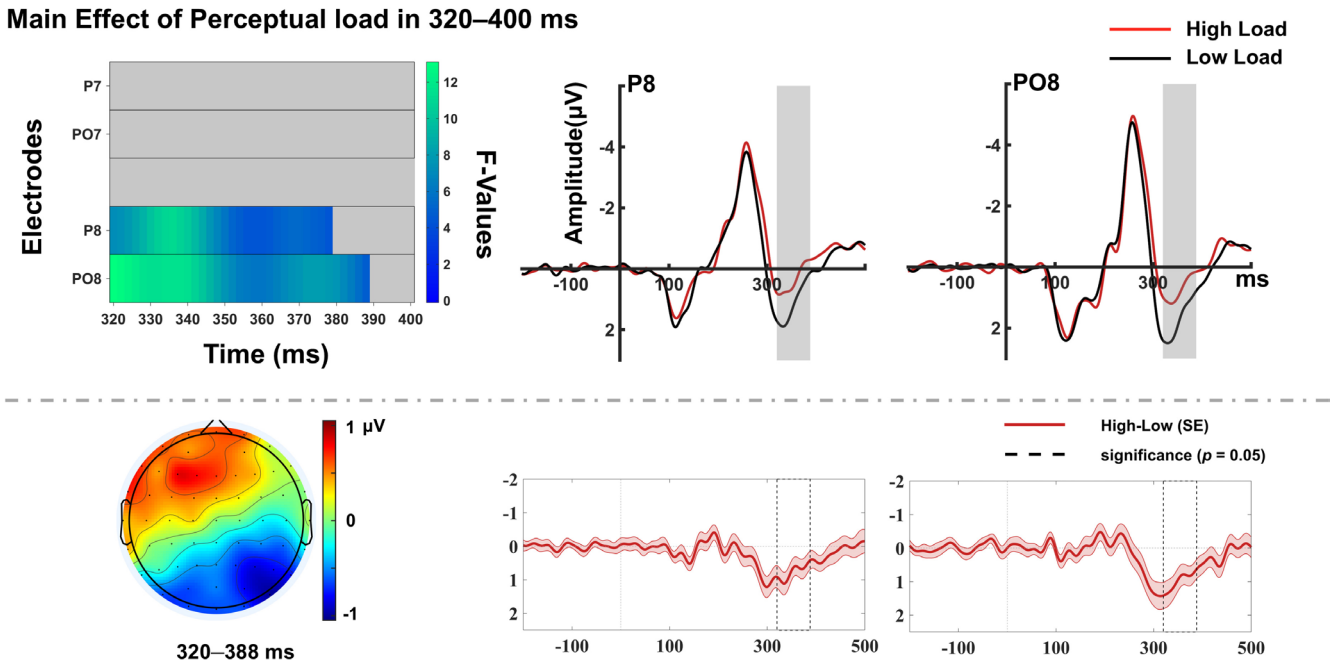


FIGURE 4 | Perceptual load main effect in the 320–400 ms time window. Grand average ERP waveforms are shown at representative electrodes (P8, PO8) for high load (red) and low load (black) conditions. The corresponding difference topographies, and difference waveforms at P8 and PO8 were displayed at the lower left the lower right panel, respectively.

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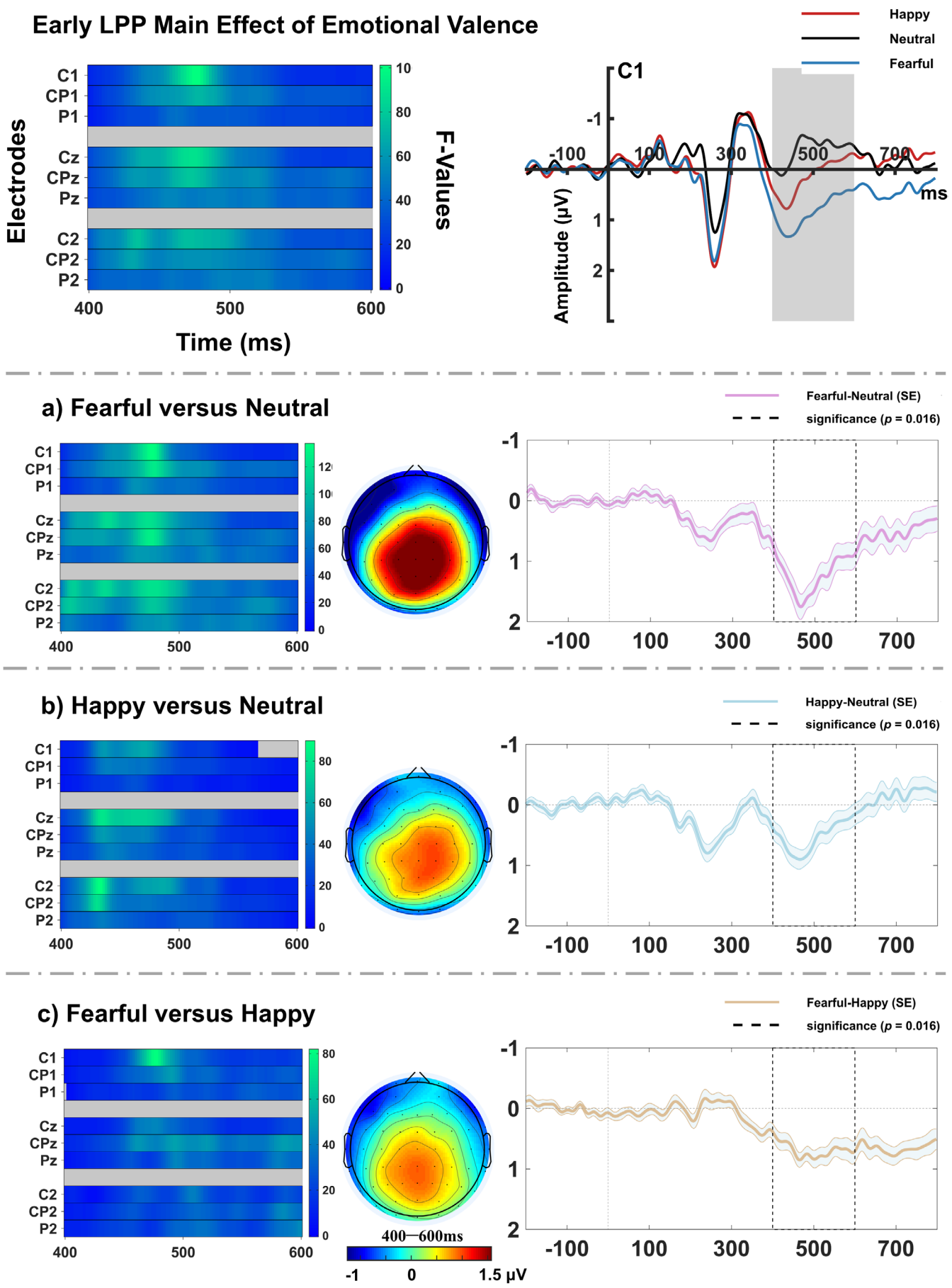


FIGURE 5 | Legend on next page.

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FIGURE 5 | Emotional valence main effect for early LPP (400–600 ms). Grand average ERP waveforms are shown at the representative electrode (C1) for fearful (blue), happy (red), and neutral (black) conditions. Follow-up ANOVAs are displayed for (a) fearful versus neutral, (b) happy versus neutral, and (c) fearful vs. happy conditions. The corresponding difference topographies and difference waveforms at C1 were displayed at the middle panel and the right panel, respectively.

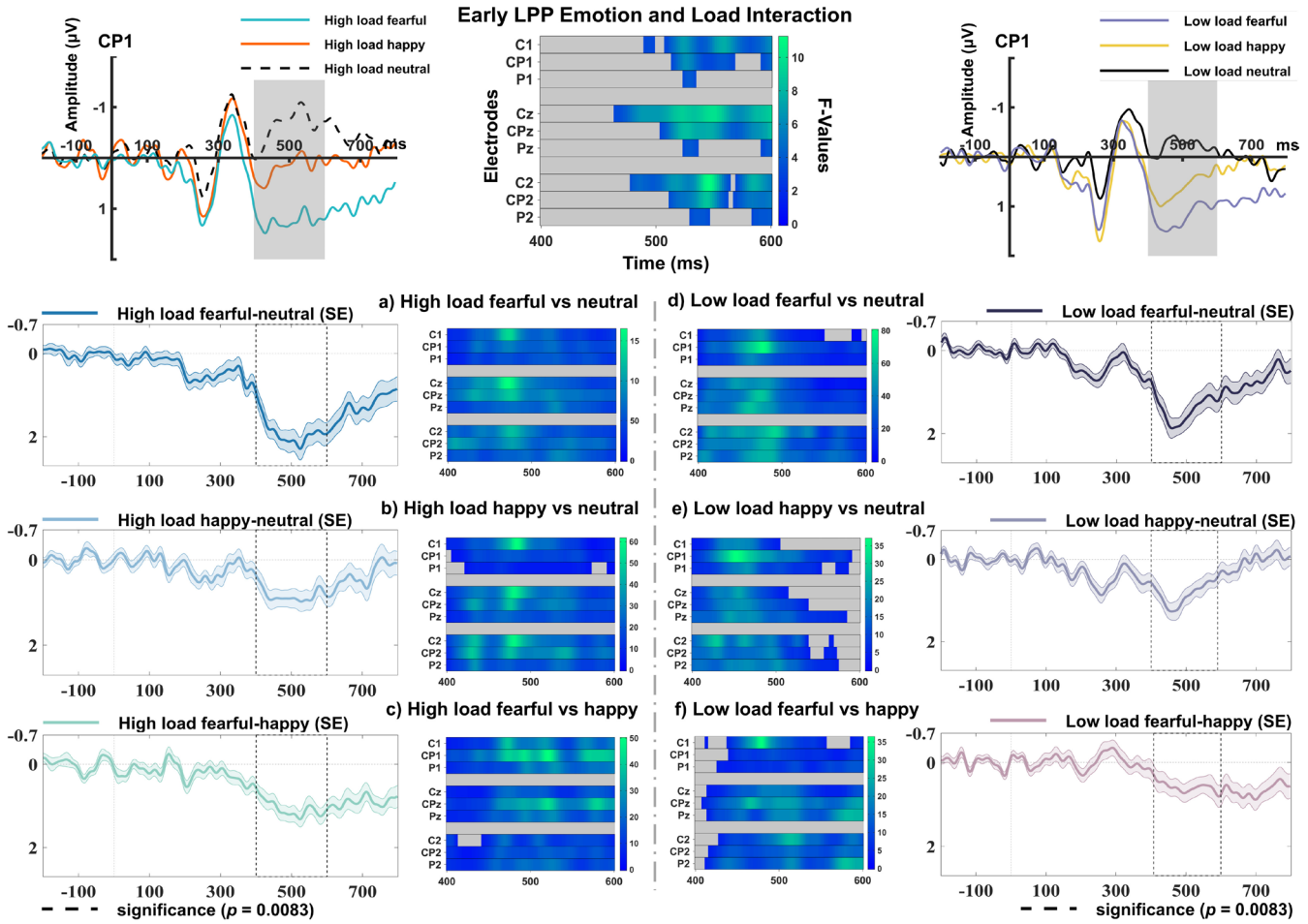


FIGURE 6 | Interaction between emotional valence and perceptual load for early LPP (400–600 ms). The left and right panels show pairwise comparisons between emotional valences under high and low load conditions, respectively, with the representative electrode (CP1) indicated. Difference waveforms at CP1 are shown alongside the grand ERP waveforms.

[$F(1, 29) = 49.89$, $p = 0.00001$]. Under the low perceptual load condition, fearful expressions were associated with enhanced amplitudes relative to neutral ones (Figure 6d), peaking at CP1 at 478 ms [$F(1, 29) = 80.31$, $p = 0.00001$]; for happy compared to neutral expressions (Figure 6e), peaking at CP1 at 452 ms [$F(1, 29) = 36.90$, $p = 0.00005$]; and for fearful compared to happy expressions (Figure 6f), peaking at C1 at 478 ms [$F(1, 29) = 36.25$, $p = 0.00006$]. The main effect of perceptual load was not significant ($ps > 0.624$).

3.2.5 | Prior Analysis (Centro-Parietal Sites; 600–800 ms)

Analysis revealed a significant main effect of emotional valence (Figure 7; Table 2). Follow-up ANOVAs revealed that the amplitude differences were observed between fearful and neutral expressions (Figure 7a), with a peak at CP2 at 632 ms [$F(1, 29) = 64.87$, $p = 0.00001$]; happy and neutral expressions (Figure 7b), peaking at CP2 at 636 ms [$F(1, 29) = 19.66$, $p = 0.00857$]; and fearful and happy expressions (Figure 7c), peaking at CPz at 604 ms [$F(1, 29) = 47.40$, $p = 0.00001$].

The interaction between emotional valence and perceptual load was significant (Figure 8; Table 2). Follow-up ANOVAs indicated that the LPP amplitude was more positive for fearful than for neutral expressions (Figure 8a), peaking at CP2 at 632 ms [$F(1, 29) = 72.81$, $p = 0.00001$]; for happy than for neutral expressions (Figure 8b), peaking at CP2 at 630 ms [$F(1, 29) = 43.46$, $p = 0.00053$]; and for fearful than for happy expressions (Figure 8c), peaking at CPz at 612 ms [$F(1, 29) = 42.67$, $p = 0.00005$] under the high load condition. As for the low perceptual load condition, the LPP amplitude was larger for fearful than for neutral expressions (Figure 8d), with a peak effect at Pz at 652 ms [$F(1, 29) = 21.39$, $p = 0.00111$]; and for fearful than for happy expressions (Figure 8f), reaching its maximal effect

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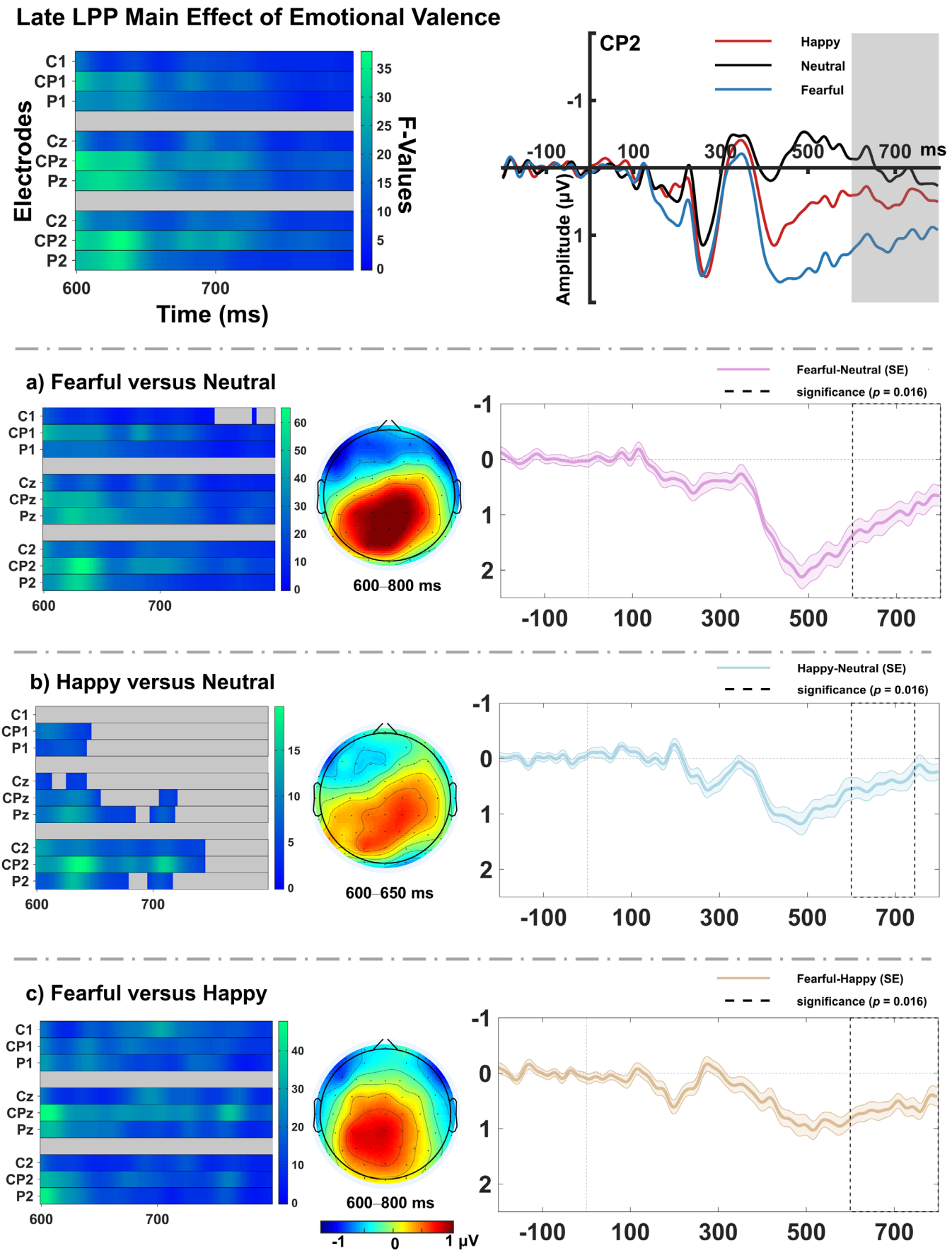


FIGURE 7 | Legend on next page.

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FIGURE 7 | Emotional valence main effect for late LPP (600–800 ms). Grand average ERP waveforms are shown at the representative electrode (CP2) for fearful (blue), happy (red), and neutral (black) conditions. Follow-up ANOVAs are displayed for (a) fearful versus neutral, (b) happy versus neutral, and (c) fearful versus happy conditions. The corresponding difference topographies and difference waveforms at CP2 were displayed at the middle panel and the right panel, respectively.

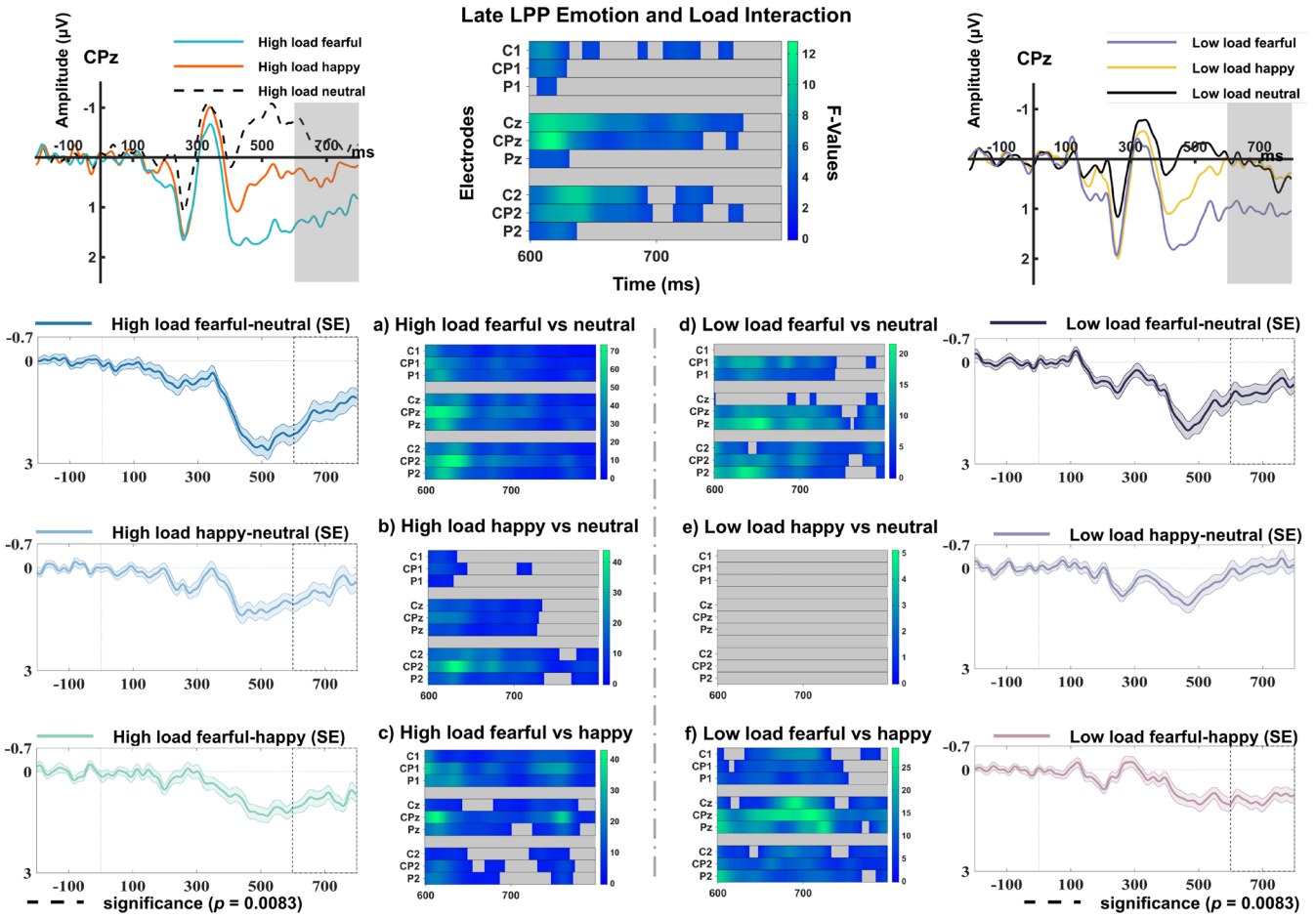


FIGURE 8 | Interaction between emotional valence and perceptual load for late LPP (600–800 ms). The left and right panels show pairwise comparisons between emotional valences under high and low load conditions, respectively, with the representative electrode (CPz) indicated. Difference waveforms at CPz are shown alongside the grand ERP waveforms.

on CPz at 708 ms [$F(1, 29) = 29.09$, $p = 0.00006$]. Comparisons between happy and neutral expressions revealed no significant differences (Figure 8e; $ps > 0.640$). The main effect of perceptual load was also nonsignificant ($ps > 0.229$).

4 | Discussion

The present study employed a modified dual-target RSVP paradigm, systematically manipulating perceptual load levels at T1 and emotional valence at T2, to examine the temporal dynamics of facial expression processing and its modulation by attentional resource availability. Emotional face processing was mainly observed in 180–280 and 400–800 ms. Consistent with our predictions, the effect of attentional resource availability on emotional face processing was limited to the late LPP (600–800 ms). Notably, the prioritized processing of fearful expressions demonstrated robustness against the constraints of reduced attentional resources. These findings contribute to a better understanding of how attentional resources interact with emotionally salient stimuli within the temporal framework of the perceptual load task.

This study demonstrates that early (180–280 ms) processing of emotional signals in faces is unaffected by perceptual load. Recent research has shown that a shorter interstimulus interval (ISI) between a forward mask and the target stimulus leads to a delayed latency and broader waveform of the N170 component (Luo et al. 2022). Moreover, current source localization revealed fusiform gyrus activation during this interval (see Supporting Informations), consistent with the face-sensitive aspects of N170 (Gao et al. 2019; Itier and Taylor 2004). Together with scalp topographies (Figure S1) and waveforms (Figure 3), the neural activity observed during this time window may primarily reflect the N170 component, which likely overlapped with earlier neural processes. A main effect of emotion was also observed during

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this time window, indicating early processing of emotional signals from faces (Mueller et al. 2025). While previous studies have reported comparable N170 responses across different emotions (Hinojosa et al. 2015; Schindler and Bublatzky 2020), we observed a significantly larger N170 for happy than fearful expressions at right occipitotemporal sites (see Figure 3c). We attribute this effect to the greater teeth exposure in happy expressions (see Figure 1c). As a key feature of the mouth, exposed teeth can capture attention to facilitate emotion recognition (Blanco et al. 2017; Horstmann et al. 2012) and also amplify the N170 amplitude irrespective of emotional valence (Brunet and Ackerman 2025). Consistent with prior research employing simultaneous stimulus presentation (Li et al. 2023; Schindler, Caldarone, et al. 2020; Schindler, Tirloni, et al. 2021), we also observed no reduction in emotional effects under high perceptual load. However, a key limitation of the simultaneous presentation design is that early parallel processing of all stimuli may occur before attentional selection, making it difficult to determine whether early emotional processing operates independently of attention (Benoni 2018; Giesbrecht et al. 2014). In contrast, the sequential paradigm used here ensures a two-stage allocation of attentional resources, providing stronger evidence for early and automatic emotion processing.

During the 320–400 ms, we observed enhanced negativity under high perceptual load conditions, consistent with previous findings suggesting a bottleneck in visual information processing (Müller-Bardorff et al. 2016; Schindler, Tirloni, et al. 2021). Given that perceptual load effects were also evident in behavioral performance, this confirms the effectiveness of the load manipulation within the dual-target RSVP paradigm. Our study included only two levels of perceptual load (87% for high load and 96% for low load), due to constraints on total experiment duration and the number of trials. However, recent findings suggest that gradually increasing perceptual difficulty may lead to nonlinear reversals of emotional effects on ERP components such as the EPN (Schindler, Bruchmann, et al. 2021). Consequently, we encourage future research to explore the effects of more complex, potentially non-linear (e.g., U-shaped) manipulations of perceptual load on the perceptual processing of emotional expressions, particularly within a temporal framework. Unexpectedly, no emotional modulation was observed during the 320–400 ms interval. Prior studies have shown that EPN emotion effects tend to emerge only when longer temporal intervals separate the perceptual load task and the emotional face stimuli, but not when the interstimulus interval (ISI) is short (e.g., 100, 300, or 600 ms) (Schindler, Bruchmann, et al. 2020; Schindler, Tirloni, et al. 2021). Since the T2 facial expressions were present at the AB period, our findings provide further evidence for a refractory window in which emotion-related modulations are suppressed.

Finally, the differentiation of emotional valences was influenced by the availability of attentional resources during more elaborative stages of processing. Previous research has shown that threat-related expressions (e.g., fearful and angry), in contrast to happy faces, consistently evoke enhanced LPP modulations across various attentional paradigms (Kühnpast et al. 2012; Schindler and Bublatzky 2020). Therefore, the LPP effects observed for happy expressions in the present study may be attributed to the task design directing attention toward emotionally salient features. Our results further demonstrated that emotional valences could be reliably distinguished during the early LPP time window, regardless of perceptual load. This likely reflects the combined influence of bottom-up emotional feedback from the amygdala and top-down modulation from frontoparietal attention networks, which together enhance visual cortical processing during the early LPP phase (Pourtois et al. 2013; Schindler and Straube 2020). Moreover, prioritized processing of fearful expressions remained stable across perceptual load levels during the late LPP window. In contrast, the differentiation between happy and neutral expressions under high load required sustained attentional engagement, whereas under low load, it was likely accomplished during the early LPP period and therefore not observed in the late LPP window. These findings align with prior evidence that the late LPP is particularly sensitive to attentional resource availability (Olofsson et al. 2008; Schupp et al. 2006; Uusberg et al. 2013). Compared to simultaneous stimulus presentation, our findings suggest that high perceptual load impairs post-perceptual processing, increasing cognitive demands for emotion discrimination (Giesbrecht et al. 2007). From an evolutionary perspective, fearful expressions serve as signals of environmental threat (Li et al. 2023; Suess et al. 2015) and effectively capture attention at both conscious and unconscious levels, as evidenced by facilitated orienting and delayed disengagement (Carlson and Mujica-Parodi 2015; Carlson and Reinke 2014; Carlson et al. 2016). Collectively, these findings indicate that processing of fearful expressions is more sustained and less reliant on available attentional resources than that of happy expressions.

This study addresses the issue of insufficient control over attentional resources in previous perceptual load research, but there remains one limitation. Previous studies (Giesbrecht et al. 2014; Giesbrecht et al. 2007) have shown that the modified dual-target RSVP task consumes attentional resources first through perceptual load, with only residual resources available for emotional face processing. To clarify the two phases of resource allocation, this study manipulated perceptual load at T1 and employed an emotion discrimination task at T2. However, previous research has consistently demonstrated task-dependent modulation of LPP components, with enhanced emotion-related amplitudes typically observed during emotion discrimination tasks relative to face detection paradigms (Aguado et al. 2019; Schindler and Straube 2020; Valdés-Conroy et al. 2014). Therefore, future studies could employ implicit tasks to better isolate and examine the impact of perceptual load on the late stages of emotional face processing.

5 | Conclusion

The present findings demonstrate that perceptual load differentially modulates early and late ERP components associated with emotional face processing, addressing when and how facial expressions depend on attentional resources. Emotional effects were observed in around 180–280 and 400–800 ms. The early emotional effects were load-independent, reflecting automatic processing of emotional expressions. A perceptual load effect emerged during 320–400 ms, suggesting a bottleneck in visual information processing. During the late LPP time window (600–800 ms), the processing of happy expressions was influenced by

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the load levels, while fearful expressions consistently enhanced LPP amplitudes regardless of attentional availability. These findings demonstrated that the processing of diverse emotional valences differs in terms of the attention resources required at the late stage. Fearful expressions, in particular, exhibit a high degree of neural automaticity, requiring minimal attentional resources for effective detection and recognition.

Author Contributions

Lan Yang: conceptualization, writing – review and editing, writing – original draft, methodology, investigation, formal analysis, visualization, software, data curation. **Shuaixia Li:** conceptualization, writing – review and editing. **Junyu Liang:** writing – review and editing. **Shuaicheng Liu:** writing – review and editing. **Jun Chen:** conceptualization, supervision, funding acquisition. **Wenbo Luo:** conceptualization, supervision, project administration, funding acquisition.

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Ethics Statement

This work was approved by the Ethics Committee of Liaoning Normal University (LL2022015) in accordance with the Declaration of Helsinki (1991). Informed consent was obtained from all subjects involved in the study.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The available data and code can be found at https://osf.io/ank6u/.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Scalp topographies of fearful, happy, and neutral expressions from 0 to 300 ms, plotted at 30-ms intervals. The 240–280 ms maps show the face-specific activation in occipito-temporal regions. **Figure S2:** The difference topographies between the three emotional valences during 0–300 ms. These maps illustrate the differences among these faces with different emotional valences are mainly evident during the time window of 180–270 ms. **Figure S3:** Source localization results of the neutral faces versus baseline difference wave during the 180–280 ms interval. The upper panel displays the activation located in the fusiform gyrus, with corresponding MNI coordinates and current density values. The middle panel shows the three-dimensional (3D) visualization of cortical activations across the whole brain. The lower panel presents the color scale representing the current density values (A/m^2). **Figure S4:** Source localization results of the happy faces versus baseline difference wave during the 180–280 ms interval. The upper panel displays the activation located in the fusiform gyrus, with corresponding MNI coordinates and current density values. The middle panel shows the three-dimensional (3D) visualization of cortical activations across the whole brain. The lower panel presents the color scale representing the current density values (A/m^2). **Figure S5:** Source localization results of the fearful faces versus baseline difference wave during the 180–280 ms interval. From top to bottom, panels depict activations in the superior temporal gyrus (STG) and fusiform gyrus, along with their corresponding MNI coordinates and current density values. A three-dimensional (3D) brain image illustrates the overall activation pattern. The lower panel shows the color scale for current density (A/m^2). **Table S1:** Brain regions showing significant activation (voxel count > 50) during the 180–280 ms interval.

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